

Health Risk Prediction Using Machine Learning Across Heterogeneous Patient Populations



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Motivation

- ***ICU decisions must be made quickly under pressure***
- ***Incorrect predictions can affect patient outcomes***
- ***Patient data is complex and varies across cases***



Problem

<i>Problem Context</i>	<i>Real-Word Issue</i>
<i>ICU mortality prediction relies on complex clinical data</i>	<i>ICU data is distributed across multiple tables</i>
<i>Multiple features must be combined for accurate modeling</i>	<i>Missing values and variability increase modeling difficulty</i>

Objective

01

Predict ICU patient mortality using classification models

02

Combine clinical features into a unified input

03

Prepare and structure data for modeling

04

Address missing values and data variability

05

Evaluate models using performance metrics

Dataset



We used the eICU Collaborative Research Database from PhysioNet.

It's a multi-center ICU dataset containing de-identified patient records.

The data includes vital signs, diagnoses, treatments, and severity scores such as APACHE.

It covers thousands of ICU stays across multiple hospitals — we used a subset of that data.

The dataset contains: 2,520 total ICU patient records 2,308 survived | 212 died

Key Challenge

The target variable was imbalanced: most patients survived, and only a small portion died.

This made it harder for the model to learn the mortality cases because it saw far fewer examples of them.

Pipeline

1

Table Selection

We selected 4 tables from the database: patient, apachePatientResult, apachePredVar, and apacheApsVar.

2

Variable Extraction

From those tables, we extracted demographics, clinical indicators, and vital signs.

3

Table Merging

We joined all tables using the patient unit stay ID as the common key.

4

Dataset Creation

We aggregated the data so each row represents one patient — one row per ICU stay.

5

Cleaning

We handled missing values through imputation and ran data consistency checks.

6

Feature Selection

We selected the most relevant clinical predictors and tested SelectKBest to compare.

7

Model Training & Evaluation

We trained models using scikit-learn pipelines and evaluated using accuracy, precision, recall, F1-score, and AUC-ROC.

Method

Approach

We treated this as a **binary classification** task — predicting whether a patient survives or dies.

Input: Processed clinical features.

Output: Patient outcome — alive or deceased.

Before modeling, we applied data preprocessing and feature engineering to prepare the input.

Models Tested

We tested eight models:

- Logistic Regression
- Random Forest
- Decision Tree
- KNN
- SVM
- Naive Bayes
- Gradient Boosting
- XGBoost

All models were built using scikit-learn and XGBoost libraries.

Tuning & Ensemble

Top 3 models selected:

Logistic Regression, XGBoost, Gradient Boosting.

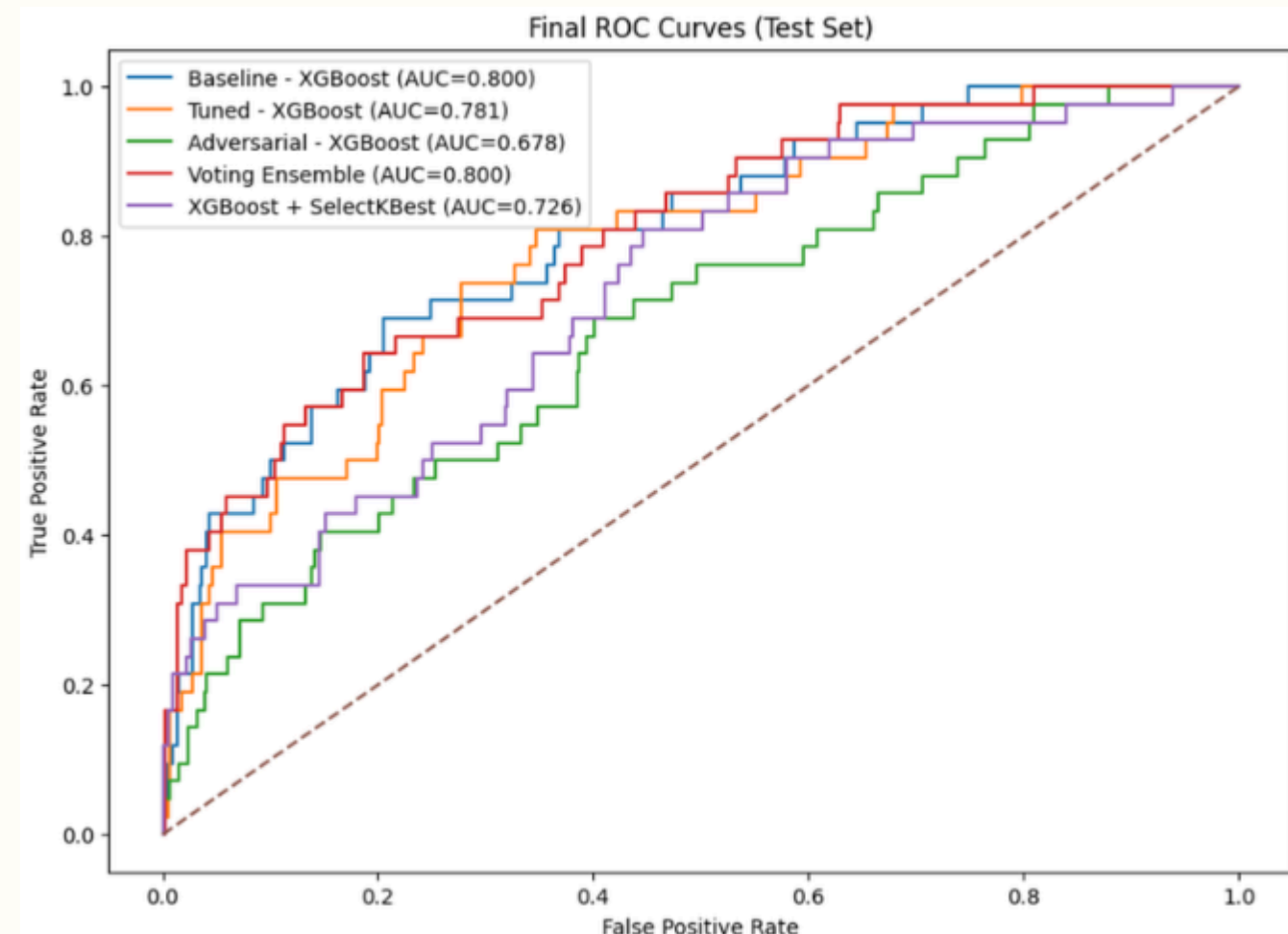
Tuning: we applied Hyperparameter tuning performed using GridSearchCV with 5-fold cross-validation

Parameters tuned:

- in XGBoost: we tuned learning rate, max depth, estimators.
- in Gradient Boosting: we tuned learning rate, estimators, depth.
- in Logistic Regression: we tuned the regularisation parameter (C).

Final model: we combined the models using Voting Ensemble — improved F1-score for the minority class.

Evaluation & Results



	Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
3	Voting Ensemble	0.892857	0.380000	0.452381	0.413043	0.800041
0	Baseline - XGBoost	0.797619	0.227273	0.595238	0.328947	0.799732
1	Tuned - XGBoost	0.861111	0.274194	0.404762	0.326923	0.780561
4	XGBoost + SelectKBest	0.815476	0.200000	0.404762	0.267717	0.725675
2	Adversarial - XGBoost	0.561508	0.125523	0.714286	0.213523	0.678056

- ***XGBoost was selected as a strong base model and used in the final ensemble***
- ***Improving performance on the minority class was challenging***
- ***Initial F1-score was very low (~0.1)***
- ***Applied hyperparameter tuning and ensemble learning to improve performance***
- ***Tested adversarial training by introducing controlled noise/perturbations to improve model robustness***
- ***Performance improved, but class imbalance remains a limitation***
- ***High accuracy with lower F1-score indicates class imbalance***

Reflection & Future Work

Reflection

- ***Class imbalance significantly impacted model performance***
- ***Accuracy was misleading because the dataset was imbalanced; F1-score was more informative because it combines precision and recall for the minority mortality class.***
- ***Real-world clinical data is complex and noisy***
- ***Model improvements required multiple strategies***

Future Work

- ***Apply class balancing techniques (e.g., SMOTE, class weighting)***
- ***Improve feature engineering for better signal extraction***
- ***Explore more advanced ensemble methods***
- ***Focus on improving minority class prediction***

Conclusion

- Developed an ICU mortality prediction pipeline using clinical data
- Compared multiple models and identified top performers
- Voting Ensemble achieved the best overall performance
- Model performance improved through tuning and model selection
- Results highlight challenges of predicting the minority class



Thank You For Listening!

Any Questions?

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